AMERICAN JOURNAL OF PHYSIOLOGY
REGULATORY, INTEGRATIVE AND COMPARATIVE PHYSIOLOGY

Am J Physiol Regul Integr Comp Physiol 328: R45–R58, 2025.
First published November 4, 2024; doi:10.1152/ajpregu.00224.2024

RESEARCH ARTICLE

Comparative analysis of acute eccentric contraction-induced changes to the skeletal muscle transcriptome in young and aged mice and humans

Jake R. Boykin,[1]† Jennifer L. Steiner,[1,6] Grant R. Laskin,[1] Michael D. Roberts,[2] Cynthia Vied,[3] Craig R. G. Willis,[4] Timothy Etheridge,[5] and Bradley S. Gordon[1,6]

[1]*Department of Health, Nutrition and Food Sciences, Florida State University, Tallahassee, Florida, United States;* [2]*School of Kinesiology, Auburn University, Auburn, Alabama, United States;* [3]*Translational Science Laboratory, Florida State University College of Medicine, Tallahassee, Florida, United States;* [4]*School of Chemistry and Biosciences, Faculty of Life Sciences, University of Bradford, Bradford, United Kingdom;* [5]*Department of Sport and Health Sciences, University of Exeter, Exeter, United Kingdom; and* [6]*Institute of Sports Sciences and Medicine, Florida State University, Tallahassee, Florida, United States*

Abstract

Adaptations to skeletal muscle following resistance exercise are due in part to changes to the skeletal muscle transcriptome. Although transcriptional changes in response to resistance exercise occur in young and aged muscles, aging alters this response. Rodent models have served great utility in defining regulatory factors that underscore the influence of mechanical load and aging on changes to skeletal muscle phenotype. Unilateral eccentric contractions in young and aged rodents are widely used to model resistance exercises in humans. However, the extent to which unilateral eccentric contractions in young and aged rodents mimic the transcriptional response in humans remains unknown. We reanalyzed two publicly available RNA sequencing datasets from young and aged mice and humans that were subjected to acute eccentric contractions to define key similarities and differences in the muscle transcriptional response following this exercise modality. The effect of aging on the number of contraction-sensitive genes, the distribution patterns of those genes into unique/common categories, and the cellular pathways associated with the differentially expressed genes (DEGs) were similar in mice and humans. However, there was little overlap between species when comparing specific contraction-sensitive DEGs within the same age group. There were strong intraspecies relationships for the common transcription factors predicted to influence the contraction-sensitive gene sets, whereas interspecies relationships were weak. Overall, these data demonstrate key similarities between mice and humans for the contraction-induced changes to the muscle transcriptome, but we posit species-specific responses exist and should be taken into consideration when attempting to translate rodent eccentric exercise models.

NEW & NOTEWORTHY Acute eccentric muscle contractions in rodents are used to model resistance exercise in young and aged humans, including changes to the muscle transcriptome. This work defines the utility of the rodent model at mimicking the transcriptional features observed in young and aged humans.

aging; gene expression; resistance exercise; transcription factors

INTRODUCTION

Skeletal muscle is important for locomotion, thermoregulation, and metabolic homeostasis (1–3), thereby influencing overall health and quality of life. Skeletal muscle phenotype can be modulated by external stimuli like resistance exercise, which promotes changes in muscle fiber size, contractile strength, and metabolic capacity (4, 5). Although both young and aged individuals benefit from resistance exercise, it is especially important for aged individuals to prevent aging-related losses in muscle mass, contractile strength, and metabolic function (6–8). Ample evidence in humans (9–18) and rodents (19–22) indicate that bouts of resistance exercise or evoked contractions significantly modulate the muscle transcriptome. The resultant gene expression patterns from several exercise bouts ultimately contribute to desirable changes in size, strength, and metabolism of the muscle (16, 23, 24).

Although resistance exercise elicits transcriptional changes in both young and aged muscles, these responses differ with age and could explain in part why adaptions to resistance exercise may be blunted in older individuals (25, 26). For instance, Raue et al. (16) reported that the skeletal muscle transcriptome was more robustly altered in younger versus older adults following acute resistance exercise. In addition, the contents of several microRNAs were differentially altered in the muscles of younger versus older adults following acute resistance exercise (27). Although these age-related transcriptional responses

†Deceased 23 September 2023.
Correspondence: B. S. Gordon (bsgordon@fsu.edu).
Submitted 13 September 2024 / Revised 16 October 2024 / Accepted 31 October 2024

0363-6119/25 Copyright © 2025 The Authors. Licensed under Creative Commons Attribution CC-BY 4.0.
Published by the American Physiological Society.

Downloaded from journals.physiology.org/journal/ajpregu (073.175.152.143) on October 8, 2026.

to resistance exercise imply divergent signaling mechanisms are operative, the upstream signals that differentially regulate contraction-induced changes in skeletal muscle gene expression (e.g., transcription factors) are less defined and are likely influenced in part by age-related changes to the content of proteins regulating gene transcription. For instance, Kurochkina et al. (28) recently showed that primary aging in humans affects the expression of ~200 nuclear-associated proteins in skeletal muscle (regulators of DNA repair, RNA processing, and transcription). Moreover, Roberts et al. (29) used deep proteomics to identify 19 transcription factors and transcriptional coregulators that were differentially expressed in a small cohort of younger versus older individuals (e.g., glucocorticoid receptor and myocyte-specific enhancer factor 2D). Several nuclear pore complex proteins were also differentially expressed between age cohorts (e.g., NUP85/93/107/133/155/160), indicating that control of transcription factor translocation from the sarcoplasm to the nucleus upon activation may be altered with age (30, 31). Thus, age-related changes to the proteome may alter the activity of the transcriptional-regulatory machinery, thereby influencing changes to the transcriptome in response to muscle contractions and ultimately the types of adaptations achieved.

Valid models to mechanistically test the effect of aging on contraction-induced transcriptional outcomes in humans are ill-defined. Much of the work elucidating the mechanistic basis for the exercise-induced signals regulating adaptations, including the effect of aging, has been performed in rodent models. The synergistic ablation model of mechanical overload has been useful in identifying critical signals regulating adaptations to resistance exercise (32). However, the synergistic ablation model has limited applicability to resistance exercise in humans, as this model lacks a recovery period following the overload stimulus as would be experienced in a human after an exercise session (33). A more physiologically relevant rodent model of resistance exercise exists, whereby the anterior crural muscles are unilaterally subjected to several sets of high force-producing eccentric contractions separated by rest periods (34), more closely resembling human resistance exercise. This model of muscle contractions also initiates several intramuscular signals occurring in the vastus lateralis muscle of humans following a bout of resistance exercise, including changes to the transcriptome (21, 35–37), potentially making it a more translatable model.

Nevertheless, biomechanical and biochemical differences between the anterior crural muscles of a rodent and the vastus lateralis of humans (the muscle typically sampled in human studies) may limit the direct translatability of the molecular response to resistance exercise in humans (38), including activation of transcriptional regulatory proteins and subsequent changes to the muscle transcriptome. In addition, the extent to which aging affects the exercise-induced transcriptional response in the anterior crural muscles of this widely used murine model compared with quadriceps muscles of humans is also largely unknown. This lack of knowledge of the congruency between the mouse model of resistance exercise and resistance exercise in humans is hindering the effective design of experiments whose data can translate to improved clinical treatments for humans. Therefore, the purpose of this study was to compare the transcriptional responses in the tibialis anterior (TA) muscle of young and aged mice produced by acute eccentric contractions to the transcriptional responses in the vastus lateralis muscle of younger and older humans following a similar protocol of acute eccentric contractions. We also sought to determine whether the upstream transcriptional regulatory signals predicted to control the transcriptional changes were similar between ages and species. To meet these objectives, we reanalyzed two publicly available RNA sequencing datasets from young and aged mice/humans subjected to acute eccentric contractions (39, 40). We hypothesized that there would be key similarities and differences between species regarding the transcriptional response and subsequent enrichment of those transcriptional findings into meaningful functional pathways. The findings of this study are important as they will guide future research practices aimed at developing clinical therapeutics that limit muscle dysfunction, particularly in aged individuals.

METHODS

Acute Eccentric Contractions in Young and Aged Male Mice

The mice used in this study and all procedures conducted have been previously described (40). Young male C57BL/6 mice ($n = 6$; 4 mo old) were purchased from Charles River (Wilmington, MA), or aged male C57BL/6 mice ($n = 4$; 24–25 mo old) were acquired from the National Institute on Aging rodent colony (NIA; Bethesda, MA). All mice were housed in a temperature (25°C) and light (12-h light/12-h dark) controlled environment within the vivarium at Florida State University and were provided standard 5001 rodent chow (LabDiet, St. Louis, MO) and water ad libitum until the day of experimentation. Animals were euthanized by cardiectomy while under deep isoflurane anesthesia. All facilities and experimental procedures were approved by the Institutional Animal Care and Use Committees of Florida State University.

The eccentric contractions protocol was conducted at the start of the mouse light cycle. Mice were subjected to unilateral high-force eccentric contractions by electrical stimulation of the sciatic nerve under deep isoflurane (3%) anesthesia (22, 41). The left thigh was shaved, two subcutaneous bipolar electrodes straddled the sciatic nerve, and the nerve was stimulated with a ~20 mA current at a frequency of 100 Hz using a constant current stimulator (Model No. 701C, Aurora Scientific, ON, Canada). Each stimulus consisted of 300 pulses, each 1 ms in duration. The entire protocol consisted of 10 sets of six stimuli with each stimulus within a set separated by a 10-s rest period, and each set separated by a 60-s rest period. Following contractions, mice received a 500 μL sc injection of warm saline before returning to their cages. During the recovery, all mice had access to water but not food. Four hours after completing the contractions protocol, the TA muscles were extracted under deep isoflurane (3%) anesthesia, snap-frozen in liquid nitrogen, and stored at −80°C until analysis.

Acute Eccentric Contractions in Young and Aged Male Humans

The human subjects used to generate the data in this study, and all procedures conducted have been previously

Downloaded from journals.physiology.org/journal/ajpregu (073.175.152.143) on October 8, 2026.

described (39). Eight young (21 ± 1 yr old, BMI 23 ± 2 kg/m^2) and eight older (70 ± 1 yr old, BMI 26 ± 1 kg/m^2) healthy untrained males volunteered for the study. A resting muscle biopsy was taken at ~0900 h during an initial visit after an overnight fast. After 96 h of having conducted no exercise, subjects again performed an overnight fast and arrived at the laboratory at ~0830 h after consuming a high-energy nutritional drink at 0700 h. The nutritional drink (Fortisip; Nutricia, Wiltshire, UK) included 150 kcal per 100 mL and had a macronutrient energy composition of 16% protein, 49% carbohydrate, and 35% fat. Subjects remained fasted thereafter until the study was completed at ~1730 h. The isolated eccentric contraction knee extension exercise protocol was performed after consuming the nutritional drink. The exercise was performed on a modified leg press designed to permit eccentric contractions in isolation within a single leg (26) with a starting knee angle of 180° for eccentric contractions. After an isolated eccentric contraction warm-up consisting of two sets of six repetitions set at 40 kg, subjects' isolated eccentric contraction one-repetition maximum (1-RM) was determined at a starting load of 50% estimated 1-RM, with subsequent attempts separated by 3 min, and the resistance increased until a controlled contraction (~3 s) through the full range of motion could no longer be performed. Immediately after eccentric 1-RM values were established, the exercise protocol was performed. The protocol comprised 7 sets of 10 repetitions (~3 s contractions) at 80% of the eccentric contraction-specific 1-RM, with 2-min rest between sets. The loads for 80% of the 1 RM in the young subjects were 211 ± 14 kg, whereas the loads for 80% of the 1 RM in the aged subjects were 155 ± 1 kg. Finally, a muscle biopsy was collected from the *m. vastus lateralis* 5 h following the cessation of the exercise protocol.

RNA Extraction and RNA Sequencing

Specific details of RNA extraction, RNA sequencing, and determination of differentially expressed genes (DEGs) from the mouse and human samples have been thoroughly described in previous work (39, 40). The raw RNA sequencing data from the mouse study are publicly available at the gene expression omnibus repository (GEO; Accession No. GSE269698), whereas the raw RNA sequencing data from the human study are publicly available in the BioProject data repository (Accession No. PRJNA509121). Notably, given that both studies used different methodologies for gene normalization (reads per kilobase per million; RPKM vs. DESeq), a direct comparison of the normalized gene counts across studies was not attempted.

Functional Gene Enrichment and Prediction of Transcription Factor Regulatory Potential

Lists of DEGs between age cohorts following exercise for each study were uploaded into the Database for Annotation, Visualization, and Integrated Discovery (DAVID; david.ncifcrf.gov) software program to define the most enriched Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways, using an unadjusted significance threshold of $P < 0.05$ (42). The top categories (max of 10), based on relevance to skeletal muscle and a significant unadjusted P value, were selected. The unadjusted P value was used, as there were few to no significant KEGG pathways if the adjustment was applied. DEG lists were also uploaded into landscape in silico deletion analysis (LISA; lisa.cistrome.org/) software to identify the transcription factors predicted to regulate each list (43). The P value associated with each transcription factor was Log transformed for the sake of visualization and used in the analysis.

Statistics

Normalized gene counts obtained from the RNA sequencing analysis were made relative to the values in the noncontracted muscle. Paired individual data points for those relative normalized gene counts are shown for select genes. The adjusted P value for those relative normalized gene counts as determined by the RNA sequencing is presented for each pair. Unpaired Student's t tests were used to compare the mean Log_2-fold changes between young and aged mice or young and aged humans. Pearson product-moment correlation analysis was used to assess relationships between the $-Log_{10}$ P values obtained from the LISA analysis. All analyses were conducted on GraphPad Prism software v. 10. Significance was established at $P \leq 0.05$ for all analyses.

RESULTS

Mice and Humans Have Common and Distinct Eccentric Contraction-Induced Changes to the Skeletal Muscle Transcriptome in Response to Aging

RNA sequencing identified 842 eccentric contraction-sensitive DEGs in the TA of young mice and 1,020 eccentric contraction-sensitive DEGs in the TA of aged mice (Fig. 1*A*). Further analysis showed that 307 DEGs were unique to young mice, 485 DEGs were unique to aged mice, and 535 DEGs were common to both (Fig. 1*A*). There were more upregulated DEGs common to young and aged mice following eccentric contractions than were unique to either age group (125 unique young vs. 426 common vs. 369 unique aged; Fig. 1*A*), whereas downregulated genes were mostly distributed evenly among the three categories (182 unique young vs. 109 common vs. 116 unique aged; Fig. 1*A*). The top KEGG pathways associated with the 307 contraction-sensitive DEGs unique to young mice contained several putative mechanically sensitive pathways related to growth, gene expression regulation, cellular reorganization, and signal transduction, including Hippo signaling, circadian rhythm, MAPK signaling, and regulation of actin cytoskeleton (Fig. 1*B*). Conversely, the top KEGG pathways associated with the 485 contraction-sensitive DEGs unique to aged mice were mostly related to metabolism/metabolic regulation, including insulin resistance, peroxisome proliferator activated receptor (PPAR) signaling, AMP-activated protein kinase (AMPK) signaling, and PI3K-Akt signaling (Fig. 1*C*). The top KEGG pathways associated with the 535 contraction-sensitive DEGs common to both young and aged mice contained putative mechanically sensitive pathways related to growth and signal transduction including Hippo signaling, focal adhesion, and MAPK signaling, as well as pathways related to inflammation (TNF signaling) and satellite cell activation (Wnt signaling and TGF-beta signaling) (Fig. 1*D*).

Downloaded from journals.physiology.org/journal/ajpregu (073.175.152.143) on October 8, 2026.

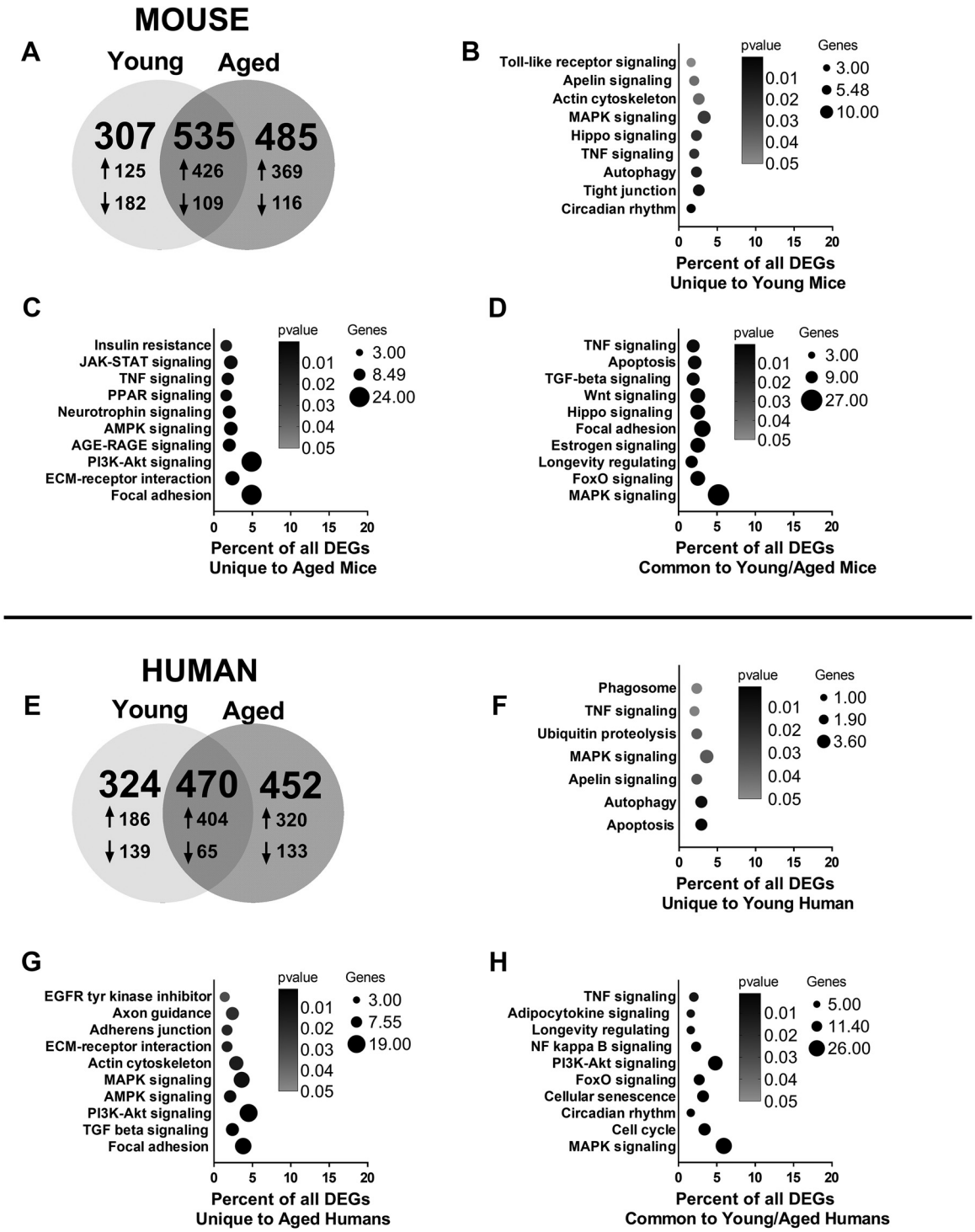


Figure 1. Mice and humans have common and distinct eccentric contraction-induced changes to the skeletal muscle transcriptome in response to aging. *A*: differentially expressed genes (DEGs) within the tibialis anterior of young mice or aged mice following acute eccentric contractions were determined by RNA sequencing analysis. DEGs were classified into categories that were distinct or common to each age group. *B–D*: KEGG pathways enriched from the DEGs common to or distinct to each mouse age group were determined using DAVID software. *E*: DEGs within the vastus lateralis muscle of young humans or aged humans following acute eccentric contractions were determined by RNA sequencing analysis. DEGs were classified into categories that were distinct or common to each age group. *F–H*: KEGG pathways enriched from the DEGs common to or distinct to each mouse age group were determined using DAVID software. DAVID, database for annotation, visualization, and integrated discovery; KEGG, Kyoto Encyclopedia of Genes and Genomes.

Downloaded from journals.physiology.org/journal/ajpregu (073.175.152.143) on October 8, 2026.

The total number of eccentric contraction sensitive DEGs in the vastus lateralis of young and aged humans, as well as the distribution pattern of the DEGs among the unique and common categories was visually similar to young and aged mice (Fig. 1*E*). There were more upregulated contraction-sensitive DEGs than downregulated DEGs in young and aged humans with the distribution patterns of those up- and downregulated DEGs among the unique and common categories being numerically similar to young and aged mice (Fig. 1*E* vs. Fig. 1*A*). The top KEGG pathways associated with the 324 contraction-sensitive DEGs unique to young humans included those related to inflammation (TNF signaling), signal transduction (MAPK signaling), and metabolism (apelin signaling and autophagy), which were also pathways unique to young mice (Fig. 1*F*). The top KEGG pathways associated with the 452 contraction-sensitive DEGs unique to aged humans included those related to extracellular matrix-cell signaling (ECM-receptor interaction), metabolic regulation (AMPK signaling and PI3K-Akt signaling), and signaling transduction (focal adhesion), which were also pathways unique to aged mice (Fig. 1*G*). The top KEGG pathways associated with the 470 contraction-sensitive DEGs common to young and aged humans included those related to inflammation (TNF signaling), metabolism (Longevity regulating and forkhead box O (FoxO) signaling), and signal transduction (MAPK signaling), which were also pathways common to young/aged mice (Fig. 1*H*). Despite some age-related similarities across species, there was some species-specific gene enrichment. For instance, only young/aged mice contained genes pertaining to Hippo signaling or Estrogen signaling, whereas only young/aged humans contained genes related to NF kappa B signaling or adipocytokine signaling (Fig. 1*D* vs. Fig. 1*H*).

Further characterization of the contraction-sensitive DEGs common to young and aged mice or common to young and aged humans also showed some similar and distinct features across species. The contraction-sensitive DEGs from aged mice had a mean Log_2-fold change that was 0.284 ± 0.366 U larger than young mice ($P < 0.0001$; Fig. 2*A*), whereas the contraction-sensitive DEGs in aged humans had a mean Log_2-fold change that was 0.0916 ± 0.463 U smaller than in young humans ($P < 0.0001$; Fig. 2*B*). In mice, the directional change in response to contractions for all DEGs common to both age groups was the same. In humans, the directional change in response to contractions for all DEGs common to both age groups was also the same except for neural cell adhesion molecule 1 (*NCAM1*), which had a directional fold change in response to contractions that differed by age (+ for young humans and − for aged humans). In mice, 14 of the contraction-sensitive DEGs common to both age groups had a Log_2-fold change that differed between age groups by more than 1 U (Young Log_2FC − Aged Log_2 FC > 1; Fig. 2*C*). Thirteen of those 14 DEGs common to both age groups with a Log_2FC differing by more than 1 U had a more pronounced directional change in aged mice compared with young mice, including the protein chaperone, heat shock protein 70 (*Hspa1a*; Fig. 2*C*). Only the fold change of inducible T-cell costimulator ligand (*Icosl*) was lower in aged mice than in young mice (Fig. 2*C*). Similar to mice, young and aged humans had 19 contraction-sensitive DEGs common to both age groups with a Log_2-fold change that differed by more than 1 U (Fig. 2*D*). In contrast to mice, 19 of the contraction-sensitive DEGs common to both age groups had a more pronounced directional change in young humans compared with aged humans, including family with sequence similarity 110 member B (*FAM110B*), which has been implicated in muscle development (44) (Fig. 2*D*).

There is Little Overlap for Specific Eccentric Contraction-Sensitive Genes between Young Mice and Young Humans despite Enrichment into Similar KEGG Pathways

Of the 842 eccentric contraction-sensitive DEGs in young mice and the 794 eccentric contraction-sensitive DEGs in young humans, only 184 were common to both species (Fig. 3*A*). One hundred sixty-one of the common contraction-sensitive DEGs were upregulated, and 23 of the common DEGs were downregulated (Fig. 3*A*). Despite the lack of congruency across species, the contraction-sensitive DEGs unique to either species were still enriched into similar KEGG pathways related to signal transduction, metabolism, and inflammation, including MAPK signaling, longevity regulating, TNF signaling, and apoptosis (Fig. 3, *B* and *C*). Those contraction-sensitive DEGs common to both young humans and young mice were enriched into KEGG pathways related to signal transduction (MAPK signaling), protein metabolism (autophagy and FoxO signaling), and gene expression regulation (circadian rhythm; Fig. 3*D*). Examples of contraction-sensitive genes unique to either species with relevance to skeletal muscle include the Hes-related family BHLH transcription factor with YRPW motif-like (*HEYL*/*Heyl*) transcription factor that influences satellite cell proliferation (45) (unique to young humans; Fig. 3*E*) and the Wnt signaling protein receptor Frizzled Class Receptor 7 (unique young mice, *FZD7*/*Fzd7*), which regulates satellite cell expansion (Fig. 3*F*) (46). An example contraction-sensitive gene common to both species with relevance to muscle is the glucose metabolic regulatory gene heparin binding EGF-like growth factor (*HBEGF*/*Hbegf*; Fig. 3*G*) (47).

Further characterization of the 184 contraction-sensitive DEGs common to both young humans and young mice showed young humans had an average Log_2-fold change in response to contractions that was 0.894 ± 0.999 U larger than young mice ($P < 0.0001$), with all DEGs from each species showing the same directional change in response to the contractions (Fig. 4*A*). There were 22 common contraction-sensitive DEGs with a Log_2-fold change differing between young humans and young mice by more than 2 U (Fig. 4*B*). Examples of those contraction-sensitive genes with relevance to skeletal muscle include the *MYC*/*Myc* transcription factor implicated in ribosome biogenesis (48), the Vestigial Like Family Member 2 (*VGLL2*/*Vgll2*) transcription factor involved with fiber type regulation (49), the inflammation regulatory transcription factor Ankyrin Repeat Domain 1 (*ANKRD1*/*Andrd1*) (50), and Salt Inducible Kinase 1 (*SIK1*/*Sik1*), which is a class II histone deacetylase (HDAC) that promotes myocyte survival (Fig. 4*B*) (51).

There is Little Overlap for Specific Eccentric Contraction-Sensitive Genes between Aged Mice and Aged Humans despite Enrichment into Similar KEGG Pathways

Of the 1,020 contraction-sensitive DEGs in aged mice and the 922 contraction-sensitive DEGs in aged humans, only 216 DEGs

Downloaded from journals.physiology.org/journal/ajpregu (073.175.152.143) on October 8, 2026.

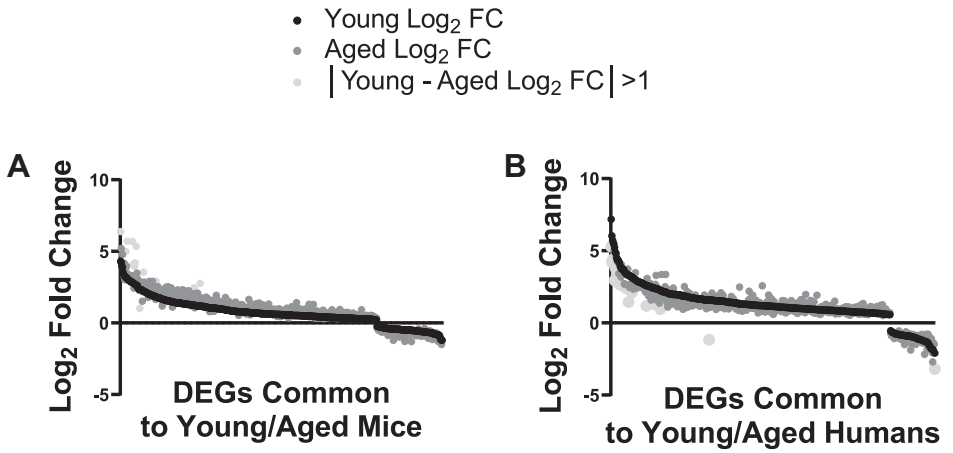


Figure 2. Characteristics associated with the differentially expressed genes (DEGs) common to young and aged mice as well as common to young and aged humans. The Log_2-fold change (FC) of DEGs common to young and aged mice (*A*) or young and aged humans (*B*). Common DEGs with an absolute Log_2 FC greater than 1 U in young and aged mice (*C*) and young and aged humans (*D*). *Example genes with known roles in muscle.

C Young Mouse Log_2 FC > 1

	Gene	Young FC	Aged FC
	Plet1	4.31	6.38
	E2f8	3.15	4.98
	Fhad1	3.09	5.71
	Hspa1b	2.80	5.68
	Arid5b	2.79	4.09
	1700101I11Rik	2.74	3.96
	Mir1938	2.70	4.19
*	*Hspa1a*	2.63	5.36
	Icosl	2.34	1.02
	Dnajb1	2.21	3.55
	Yod1	1.89	2.90
	Has1	1.80	2.90
	Gm16907	1.24	2.45
	Gm5105	1.18	2.76

D Young Human Log_2 FC > 1

	Gene	Young FC	Aged FC
	ANKRD1	7.18	5.28
	NR4A3	6.03	4.20
	TNFRSF12	5.71	4.36
	ATF3	5.54	4.42
	ACTC1	5.21	2.95
	MAFF	4.89	3.82
	CDKN1A	4.49	3.37
	OTUD1	4.36	3.16
	FAM102A	4.15	2.66
	HSPA1A	3.71	2.56
	HSPA1B	3.70	2.56
	ERRFI1	3.41	2.40
	RND3	3.30	1.45
	KLF5	3.18	2.09
	CHST15	2.60	1.18
	HSPA1L	2.42	1.31
*	*FAM110B*	2.25	0.95
	NCAM1	1.58	-1.16
	C1ORF51	-2.09	-3.19

were common to both species (Fig. 5*A*). Two hundred of the common contraction-sensitive DEGs were upregulated and 16 of the common DEGs were downregulated (Fig. 5*A*). Despite the lack of congruency between species, the contraction-sensitive DEGs unique to either species were still enriched into similar KEGG pathways related to metabolism, including insulin resistance, PI3K-Akt signaling, and AMPK signaling (Fig. 5, *B* and *C*). The contraction-sensitive DEGs common to both aged humans and aged mice were enriched into KEGG pathways related to signaling transduction, metabolism, and hormone signaling, including MAPK signaling, longevity regulating, estrogen signaling, and parathyroid hormone (Fig. 5*D*). Examples of contraction-sensitive genes unique to either species with relevance to skeletal muscle include the mitochondrial carnitine transporter Carnitine Palmitoyltransferase 1A (*CPT1A*/*Cpt1a*, unique to aged humans; Fig. 5*E*) and the insulin signaling regulatory gene Cell Death-inducing DFFA-Like Effector C (*CIDEC*/*Cidec*, unique to aged mice; Fig. 5*F*) (52). An example contraction-sensitive gene common to both species with relevance to skeletal muscle is a regulator of muscle proteolysis, Phosphodiesterase 4D (*PDE4D*/*Pde4d*; Fig. 5*G*) (53).

Further characterization of the 216 contraction-sensitive DEGs common to both aged humans and aged mice showed aged humans had an average Log_2-fold change in response to contractions that was 0.439 ± 0.890 U larger than aged mice ($P < 0.0001$) with the directional change in response to contractions being the same for all common DEGs (Fig. 6*A*). There were 17 common contraction-sensitive DEGs in aged humans and aged mice with a Log_2-fold change that differed by more than 2 U (Fig. 6*B*), with 14 of those contraction-sensitive DEGs having a greater fold change in aged humans than in aged mice. Examples of those contraction-sensitive genes with known functions in skeletal muscle include the cell cycle regulator Cyclin-Dependent Kinase Inhibitor 1A (*CDKI1A*/*Cdki1a*), the *MYC*/*Myc* transcription factor, and TNF Receptor Superfamily Member 12A (*TNFRSF12A*/*Tnfrsf12a*), which regulates proteolysis (Fig. 6*B*) (54).

There Are Strong Intraspecies Relationships for the Common Transcription Factors Predicted to Influence the Eccentric Contraction-Sensitive Gene Sets, Whereas Interspecies Relationships Are Weak

In silico analysis identified transcription factors predicted to influence the contraction-sensitive DEG sets. The *P* value associated with each identified transcription factor represents the degree of predicted influence the transcription

Downloaded from journals.physiology.org/journal/ajpregu (073.175.152.143) on October 8, 2026.

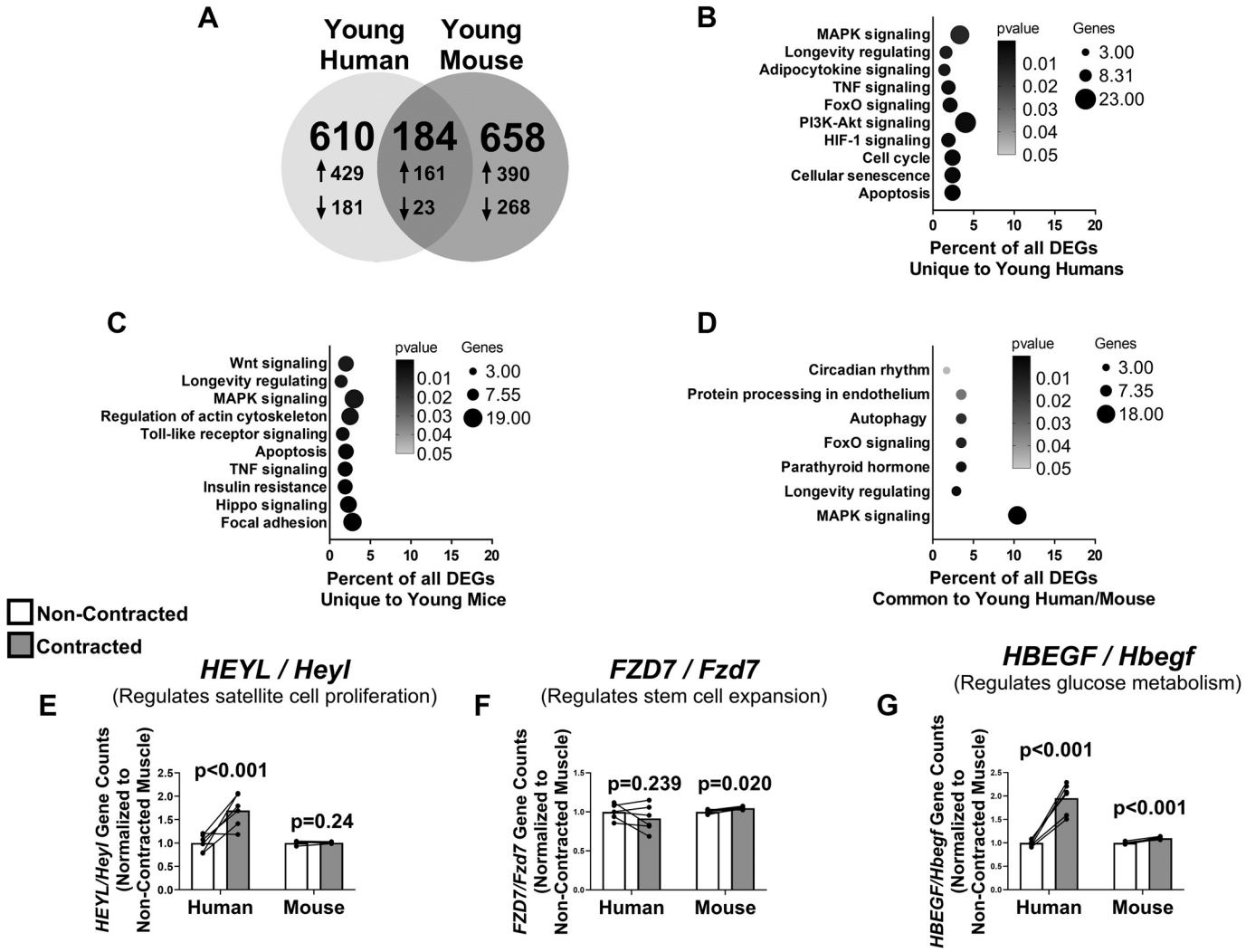


Figure 3. There is little overlap for eccentric contraction-sensitive genes specific to young mice and young humans despite enrichment into similar KEGG pathways. *A*: differentially expressed genes (DEGs) within the tibialis anterior muscle of young mice or vastus lateralis muscle of young humans were determined by RNA sequencing analysis. DEGs were classified into categories that were distinct or common to each species. *B–D*: KEGG pathways enriched from the DEGs common to or distinct to each species were determined using DAVID software. *E–G*: example genes from each classification category. The adjusted *P* value associated with each example gene was determined by RNA sequencing analysis. DAVID, database for annotation, visualization, and integrated discovery; KEGG, Kyoto Encyclopedia of Genes and Genomes.

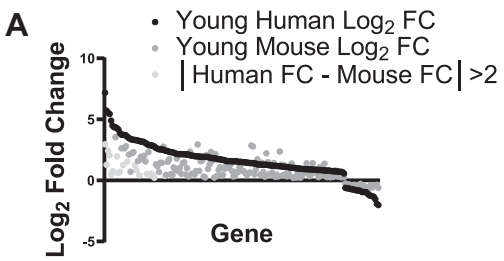
factor had over the gene set. We compared the $-\text{Log}_{10}$ *P* values for the transcription factors common between young and aged mice, between young and aged humans, between young mice and young humans, as well as between aged mice and aged humans. Using this approach, we observed very strong relationships for the common transcription factors predicted to influence the contraction-sensitive upregulated gene sets in young mice and aged mice ($r = 0.9645$, $R^2 = 0.9303$, $P < 0.0001$; Fig. 7*A*), as well as in young humans and aged humans ($r = 0.8879$, $R^2 = 0.7883$, $P < 0.0001$; Fig. 7*B*). However, the transcription factor relationships for the contraction-sensitive upregulated gene sets were quite weak when comparing young mice to young humans ($r = 0.4404$, $R^2 = 0.1939$, $P < 0.0001$; Fig. 7*C*) or aged mice to aged humans ($r = 0.4975$, $R^2 = 0.2476$, $P < 0.0001$; Fig. 7*D*).

Similar observations were noted when comparing the contraction-sensitive downregulated gene sets. There were moderate to strong relationships for common transcription factors predicted to influence the contraction-sensitive downregulated gene sets of young mice and aged mice ($r = 0.8506$, $R^2 = 0.7236$, $P < 0.0001$; Fig. 7*E*), as well as in young humans and aged humans ($r = 0.6528$, $R^2 = 0.4261$, $P < 0.0001$; Fig. 7*F*). However, the strength of the relationships was quite weak for the common transcription factors predicted to influence the contraction-sensitive downregulated gene sets for young mice to young humans ($r = 0.3544$, $R^2 = 0.1256$, $P < 0.0001$; Fig. 7*G*) or aged mice to aged humans ($r = 0.4701$, $R^2 = 0.2210$, $P < 0.0001$; Fig. 7*H*). Lists of the common transcription factors for each analysis are available as a Supplemental Table (https://DOI.org/10.6084/m9.figshare.26961568).

DISCUSSION

Several groups have shown that acute resistance exercise changes the skeletal muscle transcriptome regardless of age (9, 11–18). However, the regulatory processes underscoring

Downloaded from journals.physiology.org/journal/ajpregu (073.175.152.143) on October 8, 2026.



B

	Gene	Human FC	Mouse FC
*	*ANKRD1 / Ankrd1*	7.18	2.96
	TNFRSF12A / Tnfrsf12a	5.71	1.28
	ATF3 / Atf3	5.54	2.42
	XIRP1 / Xirp1	5.44	2.04
	MAFF / Maff	4.89	0.48
	CDKN1A / Cdkn1a	4.49	0.76
*	*SIK1 / Sik1*	4.36	0.68
	KLHL40 / Klhl40	4.27	1.97
	FAM102A / Fam102a	4.15	1.96
	RAB15 / Rab15	3.87	0.71
*	*MYC / Myc*	3.74	1.27
	HBEGF / Hbegf	3.57	1.54
	ERRFI1 / Errfi1	3.41	0.71
	RND3 / Rnd3	3.30	1.25
	ASB5 / Asb5	3.29	0.84
	FNIP2 / Fnip2	3.22	0.41
	MAST4 / Mast4	2.99	0.54
*	*VGLL2 / Vgll2*	2.89	0.43
	NUDT4 / Nudt4	2.85	0.56
	TGFB2 / Tgfb2	2.85	0.41
	EIF4E / Eif4e	2.77	0.22
	CPEB4 / Cpeb4	2.46	0.39

Figure 4. Characteristics associated with the differentially expressed genes (DEGs) common to young mice and young humans. *A*: the Log_2-fold change (FC) of DEGs common to young mice and young humans. *B*: common DEGs with an absolute Log_2 FC greater than 2 U. *Example genes with known roles in muscle.

those transcriptomic changes remain poorly defined, particularly with advanced age. There is a continued need to determine how information derived in rodent exercise models translate to humans for the successful enhancement of clinical practice. Hence, we compared the transcriptomic changes elicited in the TA muscle of the widely used eccentric contractions model of resistance exercise in young and aged mice to the changes elicited in the vastus lateralis muscle by eccentric contractions in young and aged humans. There are several important similarities between mice and humans regarding the transcriptomic responses, including the cellular programs altered by this exercise modality regardless of age. However, several key factors that appear to be species-specific were also identified. These data demonstrate that there are important similarities for the eccentric contraction-induced changes to the muscle transcriptome between mice and humans regardless of age, but that caution is warranted when making direct comparisons between the TA of mice and vastus lateralis of humans. Understanding these key similarities and differences between models will guide future research, so that data derived from preclinical models can more effectively translate to the development of clinical therapeutics in humans.

Conserved Transcriptional Responses to Eccentric Contractions and Practical Implications

The effect of aging on the number of contraction-sensitive genes, the distribution patterns of those genes into unique/common categories, as well as the KEGG pathways associated with the DEGs were similar in mice and humans. For instance, there were 307 DEGs unique to young mice and 485 unique to aged mice, which was similar to the 324 DEGs unique to young humans and 452 unique to aged humans. The KEGG pathways associated with DEGs unique to the young and aged groups were also similar between mice and humans. For example, pathways related to metabolism (Apelin signaling), signal transduction (MAPK signaling), and protein metabolism (autophagy) were enriched in young mice and young humans, whereas pathways associated with energy metabolism (AMPK signaling) and membrane-mediated signal transduction (ECM-receptor interaction and focal adhesion) were enriched in aged mice and aged humans. Regardless of species and muscle type, these data imply that aged muscle senses the mechanical and metabolic stresses of eccentric contractions differently than young muscle. The congruency between rodents and humans in this aspect means that rodents may be a valuable preclinical model to mechanistically define specific age-related changes to the muscle microenvironment that cause aging-specific transcriptional responses to eccentric contractions.

Possible changes to the molecular landscape eliciting conserved age-induced changes to the transcriptome are that aged muscle differs in the types/quantities of proteins expressed and/or metabolic capabilities (28, 29). For instance, aged muscle does not transfer forces across the muscle to the same magnitude as young muscle, partly because of conserved differences in the expression levels of force-transmitting proteins (e.g., dystrophin and sarcospan) (55, 56). The age-related differences in mechanosensing and force transfer could ultimately impact the activity of mechanically sensitive transcriptional regulatory proteins. Moreover, aged muscle is more susceptible to contraction-induced damage (55, 57–59), which likely also elicits unique transcriptional responses to the contractions. Metabolically, aerobic ATP production and intramuscular mitochondrial energetics have been reported to be impaired in aged muscle (8, 59). Thus, the energetic demand of the contractions themselves and/or the metabolic demand required for aged muscle to restore homeostasis in the postexercise recovery period may elicit a differential transcriptional response.

Species-Specific Transcriptional Responses to Eccentric Contractions and Practical Implications

Despite some of the between-species similarities reported herein, there were species-specific signatures when comparing the contraction-sensitive DEGs. For example, only 184 contraction-sensitive DEGs were conserved between young mice and young humans and only 216 contraction-sensitive DEGs conserved between aged mice and older humans. This lack of conservation between species resulted in contraction-sensitive DEGs from young/aged humans being enriched into KEGG pathways not observed in young/aged mice. Specifically, young humans showed species-specific enrichment into KEGG pathways such as Cellular senescence and PI3K-Akt signaling, whereas aged humans showed species-specific enrichment into KEGG pathways such as circadian rhythm and NF-κB signaling. There are several possible reasons for these interspecies differences that will be further discussed. Defining these differences between the rodent and human models may reduce the unfortunate failure of therapeutics (e.g., pharmaceutical) derived and tested

Downloaded from journals.physiology.org/journal/ajpregu (073.175.152.143) on October 8, 2026.

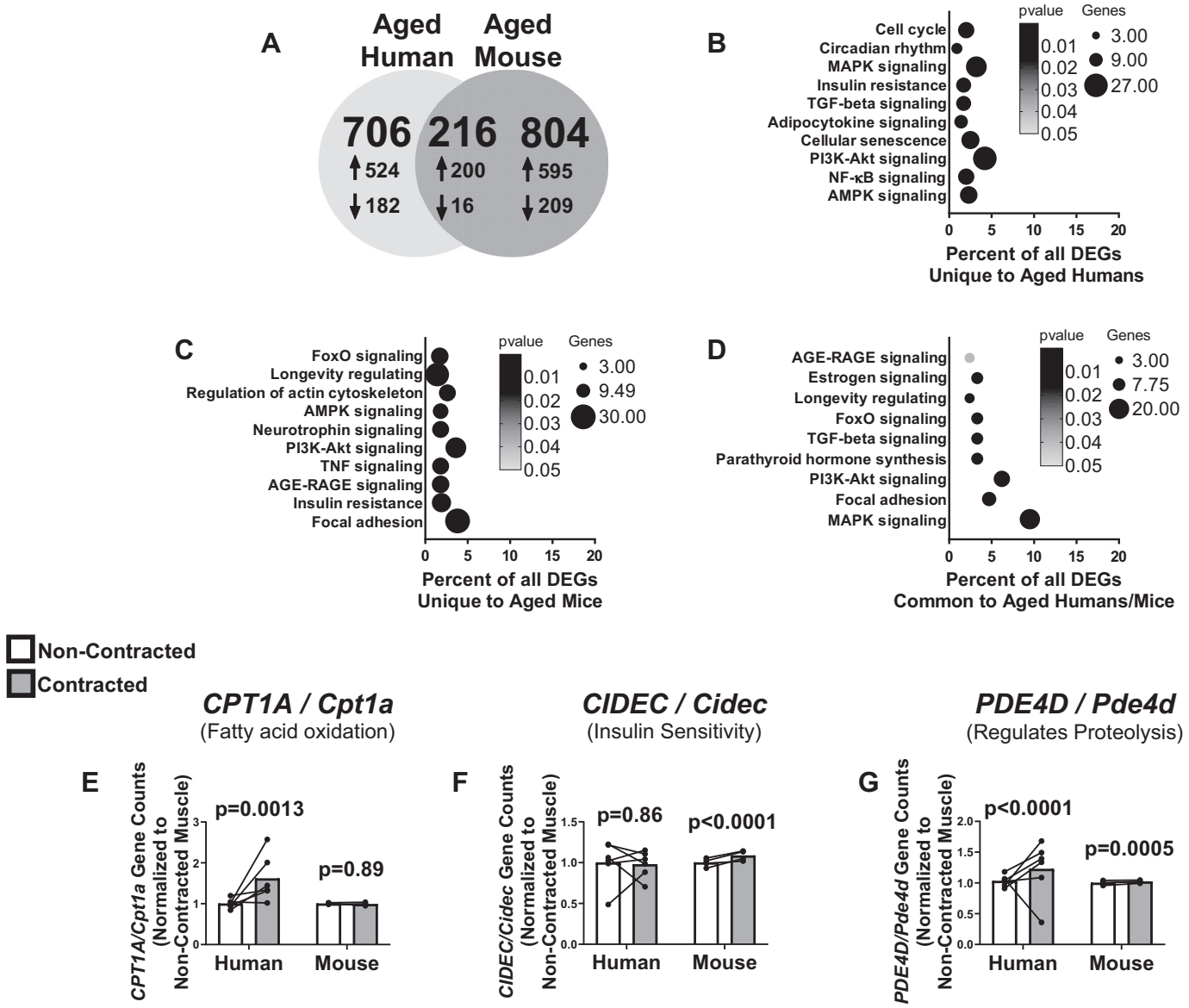


Figure 5. There is little overlap for eccentric contraction-sensitive genes specific to aged mice and aged humans despite enrichment into similar KEGG pathways. *A*: differentially expressed genes (DEGs) within the tibialis anterior muscle of aged mice or vastus lateralis muscle of age humans were determined by RNA sequencing analysis. DEGs were classified into categories that were distinct or common to each species. *B–D*: KEGG pathways enriched from the DEGs common to or distinct to each species were determined using DAVID software. *E–G*: example genes from each classification category. The adjusted *P* value associated with each example gene was determined by RNA sequencing analysis. DAVID, database for annotation, visualization, and integrated discovery; KEGG, Kyoto Encyclopedia of Genes and Genomes.\\mumnasprod\cenpro\ApplicationFiles\Journals\APS\ADVC\Vol04901\240122\ML\IProof.

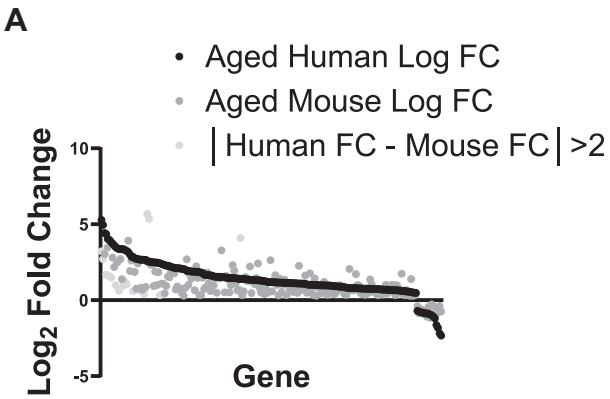
in preclinical models to effectively translate to positive clinical outcomes in humans.

Potential Reasons for Species-Specific Transcriptional Responses to Eccentric Contractions

Reasons for the lack of continuity across species within each age group could be due to several factors beginning with differences in the fiber type composition of the muscles analyzed. The tibialis anterior muscle of mice contains >95% type IIA/X/B fibers (60), whereas the vastus lateralis of humans contains roughly 50% each of type I and II fibers (61). Indeed, previous work showed that muscles containing different percentages of type I fibers (soleus) and type II fibers (plantaris) exhibit distinct magnitudes of adaptation (e.g., increase in mass) in response to synergistic ablation-induced mechanical overload (62). Moreover, fiber type-specific transcriptomic responses have been shown to occur in humans following a single bout of exercise (16). Hence, we posit that muscle fiber type differences between species are likely responsible for at least some of the species-specific signatures observed. Inducing contractions in a rodent muscle that more closely resembles the fiber type composition of the human quadriceps could yield more translational potential.

Viggars et al. (63) compared the DEG response in the TA muscles of age-matched (5 mo old) mice and rats acutely

Downloaded from journals.physiology.org/journal/ajpregu (073.175.152.143) on October 8, 2026.



B

	Gene	Human FC	Mouse FC
	ANKRD1 / Ankrd1	5.28	3.22
	XIRP1 / Xirp1	4.96	2.70
*	*TNFRSF12A / Tnfrsf12a*	4.36	1.68
	PHLDA1 / Phlda1	3.97	1.58
*	*MYC / Myc*	3.85	1.48
	THBS1 / Thbs1	3.50	0.98
*	*CDKN1A / Cdkn1a*	3.37	0.62
	IL1R1 / Il1r1	3.36	0.97
	THBD / Thbd	3.29	1.18
	ASB5 / Asb5	3.25	1.06
	TGFB2 / Tgfb2	2.85	0.60
	OSBPL8 / Osbpl8	2.67	0.65
	EIF4E / Eif4e	2.65	0.41
	HSPA1B / Hspa1b	2.56	5.68
	HSPA1A / Hspa1a	2.56	5.36
	VGLL2 / Vgll2	2.43	0.33
	ARID5B / Arid5b	1.40	4.09

Figure 6. Characteristics associated with the differentially expressed genes (DEGs) common to aged mice and aged humans. *A*: the Log_2-fold change (FC) of DEGs common to aged mice and aged humans. *B*: common DEGs with an absolute Log_2 FC greater than 2 U. *Example genes with known roles in muscle.

following a bout of eccentric contractions. Although the authors reported strong between-species concordance in the responses of genes involved with "stress responses", "collagen homeostasis", "ribosomal subunits", "autophagy," and "focal adhesion", there were several significant pathway differences between species as well. In explaining these findings, the authors posited (according to their transcriptomic data) that mice may present an increased demand for anaplerotic substrates following an exercise bout that is supplied through increased amino acid metabolism. In accordance with Viggars et al. (63), we also surmise that metabolic differences between the skeletal muscle of mice and humans could possibly contribute to the species-specific DEG signature observed herein. Rats have a nutrient metabolic profile that more closely resembles that of humans (64), including metabolic responses to acute exercise (65). Therefore, a rat model of eccentric contractions may better resemble the transcriptional responses in humans, although this possibility requires additional work.

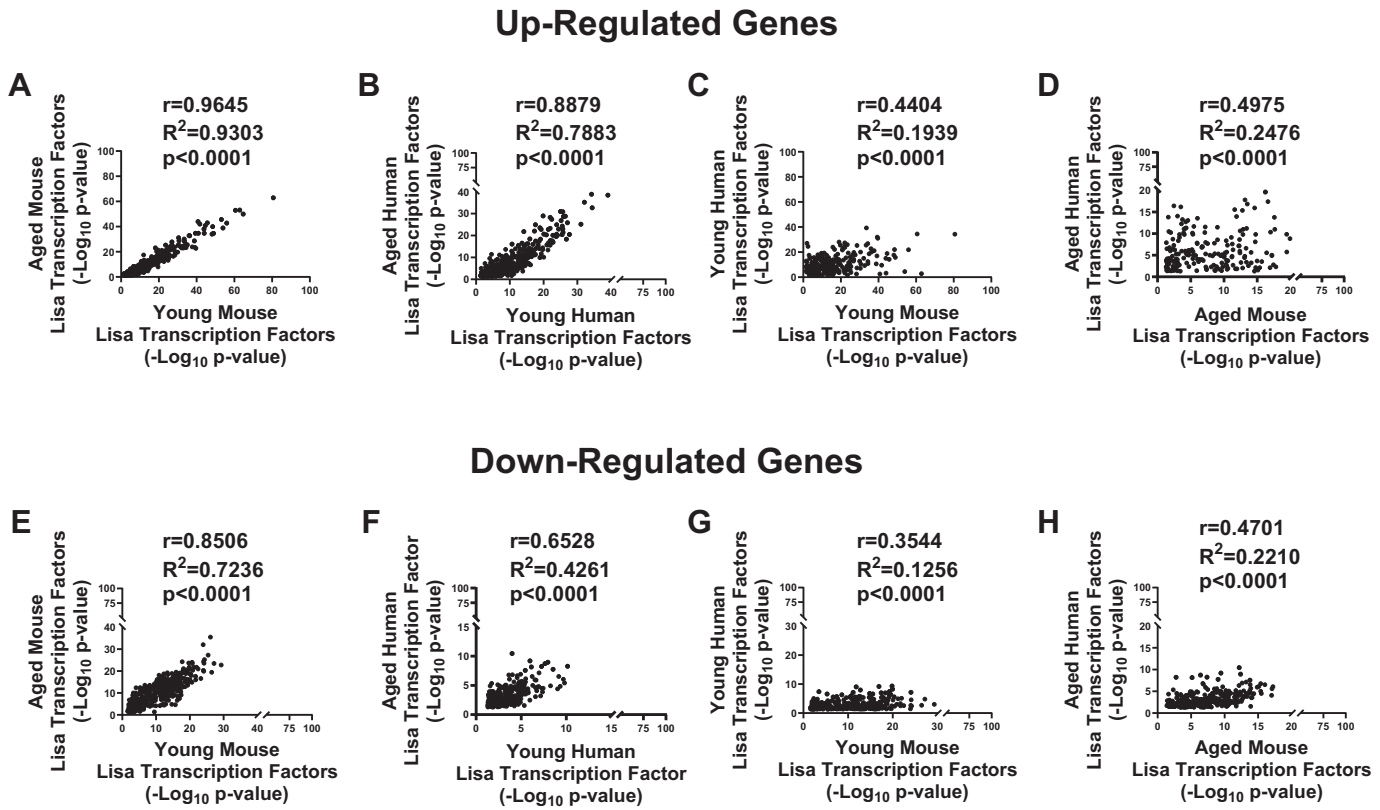


Figure 7. There are strong intraspecies relationships for the common transcription factors predicted to influence the eccentric contraction-sensitive gene sets, whereas interspecies relationships are weak. *A–H*: differentially expressed genes (DEGs) were uploaded into the landscape in silico deletion analysis (LISA; lisa.cistrome.org/) software to identify transcription factors predicted to influence the differentially expressed gene lists. The relationships between the $-Log_{10}$ *P* values for the transcription factors common to each age group or species were assessed by Pearson product-moment correlation analysis.

Downloaded from journals.physiology.org/journal/ajpregu (073.175.152.143) on October 8, 2026.

In addition, it cannot be ignored that muscles between the species were harvested at different times within the circadian cycle. Mouse muscles were harvested during the light cycle when activity/feeding levels are generally low, whereas human samples were harvested during the day when activity/feeding levels are generally high. Recent work has defined circadian rhythms within muscle (i.e., molecular clock) as a key process regulating energy production and substrate utilization pathways (66, 67), which may impact the types of genes altered in response to stimuli such as eccentric contractions. Indeed, previous work indicates that the transcriptional response to acute aerobic exercise in rodents differs whether specimens are collected in the light cycle versus the dark cycle, even if the exercise stimulus (i.e., intensity/duration) and several systemic hormonal responses are similar (68). Moreover, transcription factor regulation of gene transcription is influenced by the activity of other concurrent transcriptional regulatory events at the DNA, and the activity of those concurrent signals changes across the day (69, 70). Such a scenario may also explain in part why there were weak relationships for the common transcription factors predicted to influence the DEGs lists between mice and humans (regardless of age), despite moderate to strong relationships within a species (also regardless of age). Future work will need to assess the extent to which time of day sample collection influences the early transcriptional response to eccentric contractions between mice and humans. For instance, harvesting the mouse samples within the dark cycle may better represent the metabolic/behavioral characteristics of human muscles traditionally sampled during the daytime.

Species-specific genomic landscapes may have contributed to some of the reported interspecies differences. Humans have a genome size that is ~300 million nucleotides longer than the mouse genome, with this variation being attributed to longer introns and intergenic spaces (71). Longer intergenic spaces are likely to alter global gene expression patterns, as intergenic regions contain many *cis*-regulatory DNA elements that are functionally important for regulating gene expression (72). Alternatively, transcription factor footprints have been reported to be only ~22% conserved between mice and humans (73). Despite the difference in transcription factor footprinting, there was ~50% conservation of cross-regulatory connections between mouse and human transcription factors, and the larger scale architectural features of the mouse and human transcription factor networks were reported also to be quite similar (73). That degree of interspecies conservation may in part explain why there were several common transcription factors predicted to influence the mouse and human contraction-sensitive transcriptomes. Nonetheless, the lack of complete interspecies conservation may have resulted in the weak relationships for the degree to which common transcription factors were predicted to impart their regulation.

There were also slight differences in the exercise/muscle sampling protocols between mice and humans. Human subjects performed 70 eccentric contractions in addition to a 12-repetition warm-up and a 1 RM test, equaling >82 contractions. The mice performed just 60 contractions, meaning the degree of stimulus may have been different between the two species and therefore explain the between species differences to the magnitudes of the transcriptional responses (e.g., Figs. 4, 6, and 7). Second, there were differences in how the biopsies were collected and subsequently analyzed relative to how the mouse tissue was collected and analyzed. In the humans, baseline noncontracted muscle biopsies were taken at ~0900 h, whereas the contracted muscle biopsies were taken 5 h after the exercise session equating to ~1730 h. The comparison of gene expression in the exercised versus nonexercised leg yielded a ~8 h time differential. In mice, the contracted and noncontracted muscles were harvested at the same time, which minimizes circadian-related confounding effects with the mouse data.

Finally, the length of fast before the biopsies differed between noncontracted muscle biopsies and the contracted muscle biopsies in humans, whereas this was not a confounding factor in mice. Consumption of the nutritional drink before the eccentric contractions could have influenced the contraction-sensitive gene expression response in humans compared with the baseline biopsy that was taken after an overnight fast. Nutrient consumption alone regulates the transcriptome (74), and the combination of nutrients and contractions can further influence the transcriptome. Our laboratory showed that the consumption of nutrients before acute eccentric contractions altered the magnitude of the transcriptional response for some contraction-sensitive genes, and in some cases, reversed the directional change of the transcriptional response compared with the fasted state (22). Hence, aside from some of the species' differences mentioned in prior paragraphs, these experimental considerations between models are not trivial and are likely responsible for some of the species-specific signatures.

Conclusions

In conclusion, these data support that some of the eccentric-contraction-induced changes in the muscle transcriptome are conserved between mice and humans, including the effect of aging. However, key species differences also exist, and the role that species-specific upstream signals (i.e., transcription factors) may play in regulating contraction-sensitive genes between mice and humans requires further exploration. Critically, we posit that the molecular exploration of the differences between rodent and human exercise models is important for better understanding the strengths, limitations, and utility of each respective paradigm. Understanding the differences between preclinical and human models will help guide the design of future research aimed at improving clinical therapeutic practices.

DATA AVAILABILITY

Data will be made available upon reasonable request.

SUPPLEMENTAL MATERIAL

Supplemental table: https://DOI.org/10.6084/m9.figshare.26961568.

ACKNOWLEDGMENTS

We acknowledge the late Jake Boykin for his contribution to this study and for his friendship. The graphical abstract was created in Biorender using a department funded license.

Downloaded from journals.physiology.org/journal/ajpregu (073.175.152.143) on October 8, 2026.

GRANTS

This work was supported in part by National Institutes of Health Grant AG073445 to B.S.G.

DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

J.R.B. and B.S.G. conceived and designed research; J.R.B., C.V., T.E., and B.S.G. performed experiments; J.R.B., G.R.L., C.V., and B.S.G. analyzed data; J.R.B., J.L.S., G.R.L., M.D.R., C.V., C.R.G.W., T.E., and B.S.G. interpreted results of experiments; J.R.B. and B.S.G. prepared figures; J.R.B. and B.S.G. drafted manuscript; J.L.S., G.R.L., M.D.R., C.V., C.R.G.W., T.E., and B.S.G. edited and revised manuscript; J.L.S., G.R.L., M.D.R., C.V., C.R.G.W., T.E., and B.S.G. approved final version of manuscript.

REFERENCES

1. **Rowland LA**, **Bal NC**, **Periasamy M.** The role of skeletal-muscle-based thermogenic mechanisms in vertebrate endothermy. *Biol Rev Camb Philos Soc* 90: 1279–1297, 2015. doi:10.1111/brv.12157.
2. **Merz KE**, **Thurmond DC.** Role of skeletal muscle in insulin resistance and glucose uptake. *Compr Physiol* 10: 785–809, 2020. doi:10.1002/cphy.c190029.
3. **Brooks SV**, **Faulkner JA.** Contractile properties of skeletal muscles from young, adult and aged mice. *J Physiol* 404: 71–82, 1988. doi:10.1113/jphysiol.1988.sp017279.
4. **Ruple BA**, **Plotkin DL**, **Smith MA**, **Godwin JS**, **Sexton CL**, **McIntosh MC**, **Kontos NJ**, **Beausejour JP**, **Pagan JI**, **Rodriguez JP**, **Sheldon D**, **Knowles KS**, **Libardi CA**, **Young KC**, **Stock MS**, **Roberts MD.** The effects of resistance training to near failure on strength, hypertrophy, and motor unit adaptations in previously trained adults. *Physiol Rep* 11: e15679, 2023. doi:10.14814/phy2.15679.
5. **Hughes DC**, **Ellefsen S**, **Baar K.** Adaptations to endurance and strength training. *Cold Spring Harb Perspect Med* 8: a029769, 2018. doi:10.1101/cshperspect.a029769.
6. **Goodpaster BH**, **Park SW**, **Harris TB**, **Kritchevsky SB**, **Nevitt M**, **Schwartz AV**, **Simonsick EM**, **Tylavsky FA**, **Visser M**, **Newman AB.** The loss of skeletal muscle strength, mass, and quality in older adults: the health, aging and body composition study. *J Gerontol A Biol Sci Med Sci* 61: 1059–1064, 2006. doi:10.1093/gerona/61.10.1059.
7. **Hinkley JM**, **Cornnell HH**, **Standley RA**, **Chen EY**, **Narain NR**, **Greenwood BP**, **Bussberg V**, **Tolstikov VV**, **Kiebish MA**, **Yi F**, **Vega RB**, **Goodpaster BH**, **Coen PM.** Older adults with sarcopenia have distinct skeletal muscle phosphodiester, phosphocreatine, and phospholipid profiles. *Aging Cell* 19: e13135, 2020. doi:10.1111/acel.13135.
8. **Coen PM**, **Jubrias SA**, **Distefano G**, **Amati F**, **Mackey DC**, **Glynn NW**, **Manini TM**, **Wohlgemuth SE**, **Leeuwenburgh C**, **Cummings SR**, **Newman AB**, **Ferrucci L**, **Toledo FGS**, **Shankland E**, **Conley KE**, **Goodpaster BH.** Skeletal muscle mitochondrial energetics are associated with maximal aerobic capacity and walking speed in older adults. *J Gerontol A Biol Sci Med Sci* 68: 447–455, 2013. doi:10.1093/gerona/gls196.
9. **Damas F**, **Ugrinowitsch C**, **Libardi CA**, **Jannig PR**, **Hector AJ**, **McGlory C**, **Lixandrão ME**, **Vechin FC**, **Montenegro H**, **Tricoli V**, **Roschel H**, **Phillips SM.** Resistance training in young men induces muscle transcriptome-wide changes associated with muscle structure and metabolism refining the response to exercise-induced stress. *Eur J Appl Physiol* 118: 2607–2616, 2018. doi:10.1007/s00421-018-3984-y.
10. **Deane CS**, **Willis CRG**, **Phillips BE**, **Atherton PJ**, **Harries LW**, **Ames RM**, **Szewczyk NJ**, **Etheridge T.** Transcriptomic meta-analysis of disuse muscle atrophy vs. resistance exercise-induced hypertrophy in young and older humans. *J Cachexia Sarcopenia Muscle* 12: 629–645, 2021. doi:10.1002/jcsm.12706.
11. **Dickinson JM**, **D'Lugos AC**, **Naymik MA**, **Siniard AL**, **Wolfe AJ**, **Curtis DR**, **Huentelman MJ**, **Carroll CC.** Transcriptome response of human skeletal muscle to divergent exercise stimuli. *J Appl Physiol (1985)* 124: 1529–1540, 2018. doi:10.1152/japplphysiol.00014.2018.
12. **Liu D**, **Sartor MA**, **Nader GA**, **Gutmann L**, **Treutelaar MK**, **Pistilli EE**, **Iglayreger HB**, **Burant CF**, **Hoffman EP**, **Gordon PM.** Skeletal muscle gene expression in response to resistance exercise: sex specific regulation. *BMC Genomics* 11: 659, 2010. doi:10.1186/1471-2164-11-659.
13. **Murton AJ**, **Billeter R**, **Stephens FB**, **Des Etages SG**, **Graber F**, **Hill RJ**, **Marimuthu K**, **Greenhaff PL.** Transient transcriptional events in human skeletal muscle at the outset of concentric resistance exercise training. *J Appl Physiol (1985)* 116: 113–125, 2014. doi:10.1152/japplphysiol.00426.2013.
14. **Nader GA**, **von Walden F**, **Liu C**, **Lindvall J**, **Gutmann L**, **Pistilli EE**, **Gordon PM.** Resistance exercise training modulates acute gene expression during human skeletal muscle hypertrophy. *J Appl Physiol (1985)* 116: 693–702, 2014. doi:10.1152/japplphysiol.01366.2013.
15. **Pillon NJ**, **Gabriel BM**, **Dollet L**, **Smith JAB**, **Sardón Puig L**, **Botella J**, **Bishop DJ**, **Krook A**, **Zierath JR.** Transcriptomic profiling of skeletal muscle adaptations to exercise and inactivity. *Nat Commun* 11: 470, 2020. doi:10.1038/s41467-019-13869-w.
16. **Raue U**, **Trappe TA**, **Estrem ST**, **Qian HR**, **Helvering LM**, **Smith RC**, **Trappe S.** Transcriptome signature of resistance exercise adaptations: mixed muscle and fiber type specific profiles in young and old adults. *J Appl Physiol (1985)* 112: 1625–1636, 2012. doi:10.1152/japplphysiol.00435.2011.
17. **Sexton CL**, **Godwin JS**, **McIntosh MC**, **Ruple BA**, **Osburn SC**, **Hollingsworth BR**, **Kontos NJ**, **Agostinelli PJ**, **Kavazis AN**, **Ziegenfuss TN**, **Lopez HL**, **Smith R**, **Young KC**, **Dwaraka VB**, **Frugé AD**, **Mobley CB**, **Sharples AP**, **Roberts MD.** Skeletal muscle DNA methylation and mRNA responses to a bout of higher versus lower load resistance exercise in previously trained men. *Cells* 12: 263, 2023. doi:10.3390/cells12020263.
18. **Turner DC**, **Seaborne RA**, **Sharples AP.** Comparative transcriptome and methylome analysis in human skeletal muscle anabolism, hypertrophy and epigenetic memory. *Sci Rep* 9: 4251, 2019. doi:10.1038/s41598-019-40787-0.
19. **Viggars MR**, **Sutherland H**, **Lanmüller H**, **Schmoll M**, **Bijak M**, **Jarvis JC.** Adaptation of the transcriptional response to resistance exercise over 4 weeks of daily training. *FASEB J* 37: e22686, 2023. doi:10.1096/fj.202201418R.
20. **Barash IA**, **Mathew L**, **Ryan AF**, **Chen J**, **Lieber RL.** Rapid muscle-specific gene expression changes after a single bout of eccentric contractions in the mouse. *Am J Physiol Cell Physiol* 286: C355–C364, 2004. doi:10.1152/ajpcell.00211.2003.
21. **Kano R**, **Kusano T**, **Takeda R**, **Shirakawa H**, **Poole DC**, **Kano Y**, **Hoshino D.** Eccentric contraction increases hydrogen peroxide levels and alters gene expression through Nox2 in skeletal muscle of male mice. *J Appl Physiol (1985)* 137: 778–788, 2024 doi:10.1152/japplphysiol.00335.2024.
22. **Laskin GR**, **Gordon BS.** The influence of nutrients on mechanical overload-induced changes to skeletal muscle mRNA content. *Physiol Genomics* 54: 360–369, 2022. doi:10.1152/physiolgenomics.00075.2022.
23. **Sartori R**, **Romanello V**, **Sandri M.** Mechanisms of muscle atrophy and hypertrophy: implications in health and disease. *Nat Commun* 12: 330, 2021. doi:10.1038/s41467-020-20123-1.
24. **Roberts MD**, **McCarthy JJ**, **Hornberger TA**, **Phillips SM**, **Mackey AL**, **Nader GA**, **Boppart MD**, **Kavazis AN**, **Reidy PT**, **Ogasawara R**, **Libardi CA**, **Ugrinowitsch C**, **Booth FW**, **Esser KA.** Mechanisms of mechanical overload-induced skeletal muscle hypertrophy: current understanding and future directions. *Physiol Rev* 103: 2679–2757, 2023. doi:10.1152/physrev.00039.2022.
25. **Brook MS**, **Wilkinson DJ**, **Mitchell WK**, **Lund JN**, **Phillips BE**, **Szewczyk NJ**, **Greenhaff PL**, **Smith K**, **Atherton PJ.** Synchronous deficits in cumulative muscle protein synthesis and ribosomal biogenesis underlie age-related anabolic resistance to exercise in humans. *J Physiol* 594: 7399–7417, 2016. doi:10.1113/JP272857.
26. **Phillips BE**, **Williams JP**, **Greenhaff PL**, **Smith K**, **Atherton PJ.** Physiological adaptations to resistance exercise as a function of age. *JCI Insight* 2: e95581, 2017. doi:10.1172/jci.insight.95581.
27. **Drummond MJ**, **McCarthy JJ**, **Fry CS**, **Esser KA**, **Rasmussen BB.** Aging differentially affects human skeletal muscle microRNA expression at rest and after an anabolic stimulus of resistance exercise and

Downloaded from journals.physiology.org/journal/ajpregu (073.175.152.143) on October 8, 2026.

essential amino acids. *Am J Physiol Endocrinol Physiol* 295: E1333–E1340, 2008. doi:10.1152/ajpendo.90562.2008.

28. **Kurochkina NS**, **Orlova MA**, **Vigovskiy MA**, **Zgoda VG**, **Vepkhvadze TF**, **Vavilov NE**, **Makhnovskii PA**, **Grigorieva OA**, **Boroday YR**, **Philippov VV**, **Lednev EM**, **Efimenko AY**, **Popov DV.** Age-related changes in human skeletal muscle transcriptome and proteome are more affected by chronic inflammation and physical inactivity than primary aging. *Aging Cell* 23: e14098, 2024. doi:10.1111/acel.14098.
29. **Roberts MD**, **Ruple BA**, **Godwin JS**, **McIntosh MC**, **Chen S-Y**, **Kontos NJ**, **Agyin-Birikorang A**, **Michel M**, **Plotkin DL**, **Mattingly ML**, **Mobley B**, **Ziegenfuss TN**, **Fruge AD**, **Kavazis AN.** A novel deep proteomic approach in human skeletal muscle unveils distinct molecular signatures affected by aging and resistance training. *Aging (Albany NY)* 16: 6631–6651, 2024. doi:10.18632/aging.205751.
30. **Raices M**, **Bukata L**, **Sakuma S**, **Borlido J**, **Hernandez LS**, **Hart DO**, **D'Angelo MA.** Nuclear pores regulate muscle development and maintenance by assembling a localized Mef2C complex. *Dev Cell* 41: 540–554.e7, 2017. doi:10.1016/j.devcel.2017.05.007.
31. **Taylor-Weiner H**, **Grigsby CL**, **Ferreira DMS**, **Dias JM**, **Stevens MM**, **Ruas JL**, **Teixeira AI.** Modeling the transport of nuclear proteins along single skeletal muscle cells. *Proc Natl Acad Sci USA* 117: 2978–2986, 2020. doi:10.1073/pnas.1919600117.
32. **Burke BI**, **Ismaeel A**, **McCarthy JJ.** The utility of the rodent synergist ablation model in identifying molecular and cellular mechanisms of skeletal muscle hypertrophy. *Am J Physiol Cell Physiol* 327: C601–C606, 2024. doi:10.1152/ajpcell.00362.2024.
33. **Roberts MD**, **Hornberger TA**, **Phillips SM.** The utility-and limitations-of the rodent synergist ablation model in examining mechanisms of skeletal muscle hypertrophy. *Am J Physiol Cell Physiol* 327: C607–C613, 2024. doi:10.1152/ajpcell.00405.2024.
34. **Wong TS**, **Booth FW.** Protein metabolism in rat tibialis anterior muscle after stimulated chronic eccentric exercise. *J Appl Physiol (1985)* 69: 1718–1724, 1990. doi:10.1152/jappl.1990.69.5.1718.
35. **Jacobs BL**, **You JS**, **Frey JW**, **Goodman CA**, **Gundermann DM**, **Hornberger TA.** Eccentric contractions increase the phosphorylation of tuberous sclerosis complex-2 (TSC2) and alter the targeting of TSC2 and the mechanistic target of rapamycin to the lysosome. *J Physiol* 591: 4611–4620, 2013. doi:10.1113/jphysiol.2013.256339.
36. **O'Neil TK**, **Duffy LR**, **Frey JW**, **Hornberger TA.** The role of phosphoinositide 3-kinase and phosphatidic acid in the regulation of mammalian target of rapamycin following eccentric contractions. *J Physiol* 587: 3691–3701, 2009. doi:10.1113/jphysiol.2009.173609.
37. **Baar K**, **Esser K.** Phosphorylation of p70(S6k) correlates with increased skeletal muscle mass following resistance exercise. *Am J Physiol Cell Physiol* 276: C120–C127, 1999. doi:10.1152/ajpcell.1999.276.1.C120.
38. **Hu X**, **Charles JP**, **Akay T**, **Hutchinson JR**, **Blemker SS.** Are mice good models for human neuromuscular disease? Comparing muscle excursions in walking between mice and humans. *Skelet Muscle* 7: 26, 2017. doi:10.1186/s13395-017-0143-9.
39. **Deane CS**, **Ames RM**, **Phillips BE**, **Weedon MN**, **Willis CRG**, **Boereboom C**, **Abdulla H**, **Bukhari SSI**, **Lund JN**, **Williams JP**, **Wilkinson DJ**, **Smith K**, **Gallagher IJ**, **Kadi F**, **Szewczyk NJ**, **Atherton PJ**, **Etheridge T.** The acute transcriptional response to resistance exercise: impact of age and contraction mode. *Aging (Albany NY)* 11: 2111–2126, 2019. doi:10.18632/aging.101904.
40. **Laskin GR**, **Waddell DS**, **Vied C**, **Gordon BS.** Contractile regulation of the glucocorticoid-sensitive transcriptome in young and aged skeletal muscle. *Am J Physiol Endocrinol Physiol* 327: E636–E652, 2024. doi:10.1152/ajpendo.00223.2024.
41. **Laskin GR**, **Cabrera AR**, **Greene NP**, **Tomko RJ**, **Vied C**, **Gordon BS.** The mechanosensitive gene arrestin domain containing 2 regulates myotube diameter with direct implications for disuse atrophy with aging. *Am J Physiol Cell Physiol* 326: C768–C783, 2024. doi:10.1152/ajpcell.00444.2023.
42. **Huang DW**, **Sherman BT**, **Tan Q**, **Collins JR**, **Alvord WG**, **Roayaei J**, **Stephens R**, **Baseler MW**, **Lane HC**, **Lempicki RA.** The DAVID Gene Functional Classification Tool: a novel biological module-centric algorithm to functionally analyze large gene lists. *Genome Biol* 8: R183, 2007. doi:10.1186/gb-2007-8-9-r183.
43. **Qin Q**, **Fan J**, **Zheng R**, **Wan C**, **Mei S**, **Wu Q**, **Sun H**, **Brown M**, **Zhang J**, **Meyer CA**, **Liu XS.** Lisa: inferring transcriptional regulators through integrative modeling of public chromatin accessibility and ChIP-seq data. *Genome Biol* 21: 32, 2020. doi:10.1186/s13059-020-1934-6.
44. **Mohammadabadi M**, **Bordbar F**, **Jensen J**, **Du M**, **Guo W.** Key genes regulating skeletal muscle development and growth in farm animals. *Animals (Basel)* 11: 835, 2021. doi:10.3390/ani11030835.
45. **Fukuda S**, **Kaneshige A**, **Kaji T**, **Noguchi YT**, **Takemoto Y**, **Zhang L**, **Tsujikawa K**, **Kokubo H**, **Uezumi A**, **Maehara K**, **Harada A**, **Ohkawa Y**, **Fukada SI.** Sustained expression of HeyL is critical for the proliferation of muscle stem cells in overloaded muscle. *eLife* 8: e48284, 2019. doi:10.7554/eLife.48284.
46. **Bentzinger CF**, **von Maltzahn J**, **Dumont NA**, **Stark DA**, **Wang YX**, **Nhan K**, **Frenette J**, **Cornelison DD**, **Rudnicki MA.** Wnt7a stimulates myogenic stem cell motility and engraftment resulting in improved muscle strength. *J Cell Biol* 205: 97–111, 2014. doi:10.1083/jcb.201310035.
47. **Fukatsu Y**, **Noguchi T**, **Hosooka T**, **Ogura T**, **Kotani K**, **Abe T**, **Shibakusa T**, **Inoue K**, **Sakai M**, **Tobimatsu K**, **Inagaki K**, **Yoshioka T**, **Matsuo M**, **Nakae J**, **Matsuki Y**, **Hiramatsu R**, **Kaku K**, **Okamura H**, **Fushiki T**, **Kasuga M.** Muscle-specific overexpression of heparin-binding epidermal growth factor-like growth factor increases peripheral glucose disposal and insulin sensitivity. *Endocrinology* 150: 2683–2691, 2009. doi:10.1210/en.2008-1647.
48. **Mori T**, **Ato S**, **Knudsen JR**, **Henriquez-Olguin C**, **Li Z**, **Wakabayashi K**, **Suginohara T**, **Higashida K**, **Tamura Y**, **Nakazato K**, **Jensen TE**, **Ogasawara R.** c-Myc overexpression increases ribosome biogenesis and protein synthesis independent of mTORC1 activation in mouse skeletal muscle. *Am J Physiol Endocrinol Physiol* 321: E551–E559, 2021. doi:10.1152/ajpendo.00164.2021.
49. **Honda M**, **Tsuchimochi H**, **Hitachi K**, **Ohno S.** Transcriptional cofactor Vgll2 is required for functional adaptations of skeletal muscle induced by chronic overload. *J Cell Physiol* 234: 15809–15824, 2019. doi:10.1002/jcp.28239.
50. **Liu XH**, **Bauman WA**, **Cardozo C.** ANKRD1 modulates inflammatory responses in C2C12 myoblasts through feedback inhibition of NF-κB signaling activity. *Biochem Biophys Res Commun* 464: 208–213, 2015. doi:10.1016/j.bbrc.2015.06.118.
51. **Berdeaux R**, **Goebel N**, **Banaszynski L**, **Takemori H**, **Wandless T**, **Shelton GD**, **Montminy M.** SIK1 is a class II HDAC kinase that promotes survival of skeletal myocytes. *Nat Med* 13: 597–603, 2007. doi:10.1038/nm1573.
52. **Gupta A**, **Balakrishnan B**, **Karki S**, **Slayton M**, **Jash S**, **Banerjee S**, **Grahn THM**, **Jambunathan S**, **Disney S**, **Hussein H**, **Kong D**, **Lowell BB**, **Natarajan P**, **Reddy UK**, **Gokce N**, **Sharma VM**, **Puri V.** Human CIDEC transgene improves lipid metabolism and protects against high-fat diet-induced glucose intolerance in mice. *J Biol Chem* 298: 102347, 2022. doi:10.1016/j.jbc.2022.102347.
53. **Arcaro CA**, **Assis RP**, **Oliveira JO**, **Zanon NM**, **Paula-Gomes S**, **Navegantes LCC**, **Kettelhut IC**, **Brunetti IL**, **Baviera AM.** Phosphodiesterase 4 inhibition restrains muscle proteolysis in diabetic rats by activating PKA and EPAC/Akt effectors and inhibiting FoxO factors. *Life Sci* 278: 119563, 2021. doi:10.1016/j.lfs.2021.119563.
54. **Bhatnagar S**, **Kumar A.** The TWEAK-Fn14 system: breaking the silence of cytokine-induced skeletal muscle wasting. *Curr Mol Med* 12: 3–13, 2012. doi:10.2174/156652412798376107.
55. **Hughes DC**, **Marcotte GR**, **Marshall AG**, **West DWD**, **Baehr LM**, **Wallace MA**, **Saleh PM**, **Bodine SC**, **Baar K.** Age-related differences in dystrophin: impact on force transfer proteins, membrane integrity, and neuromuscular junction stability. *J Gerontol A Biol Sci Med Sci* 72: 640–648, 2017. doi:10.1093/gerona/glw109.
56. **Ubaida-Mohien C**, **Lyashkov A**, **Gonzalez-Freire M**, **Tharakan R**, **Shardell M**, **Moaddel R**, **Semba RD**, **Chia CW**, **Gorospe M**, **Sen R**, **Ferrucci L.** Discovery proteomics in aging human skeletal muscle finds change in spliceosome, immunity, proteostasis and mitochondria. *eLife* 8: e49874, 2019. doi:10.7554/eLife.49874.
57. **Dedrick ME**, **Clarkson PM.** The effects of eccentric exercise on motor performance in young and older women. *Eur J Appl Physiol Occup Physiol* 60: 183–186, 1990. doi:10.1007/BF00839156.
58. **Ramaswamy KS**, **Palmer ML**, **van der Meulen JH**, **Renoux A**, **Kostrominova TY**, **Michele DE**, **Faulkner JA.** Lateral transmission of force is impaired in skeletal muscles of dystrophic mice and very old rats. *J Physiol* 589: 1195–1208, 2011. doi:10.1113/jphysiol.2010.201921.
59. **Manfredi TG**, **Fielding RA**, **O'Reilly KP**, **Meredith CN**, **Lee HY**, **Evans WJ.** Plasma creatine kinase activity and exercise-induced muscle damage in older men. *Med Sci Sports Exerc* 23: 1028–1034, 1991.

Downloaded from journals.physiology.org/journal/ajpregu (073.175.152.143) on October 8, 2026.

60. **Tyagi S**, **Beqollari D**, **Lee CS**, **Walker LA**, **Bannister RA.** Semi-automated analysis of mouse skeletal muscle morphology and fiber-type composition. *J Vis Exp* 126: 56024, 2017. doi:10.3791/56024.
61. **Staron RS**, **Hagerman FC**, **Hikida RS**, **Murray TF**, **Hostler DP**, **Crill MT**, **Ragg KE**, **Toma K.** Fiber type composition of the vastus lateralis muscle of young men and women. *J Histochem Cytochem Off J Histochem Soc* 48: 623–629, 2000. doi:10.1177/002215540004800506.
62. **Roberts MD**, **Mobley CB**, **Vann CG**, **Haun CT**, **Schoenfeld BJ**, **Young KC**, **Kavazis AN.** Synergist ablation-induced hypertrophy occurs more rapidly in the plantaris than soleus muscle in rats due to different molecular mechanisms. *Am J Physiol Regul Integr Comp Physiol* 318: R360–R368, 2020. doi:10.1152/ajpregu.00304.2019.
63. **Viggars MR**, **Sutherland H**, **Cardozo CP**, **Jarvis JC.** Conserved and species-specific transcriptional responses to daily programmed resistance exercise in rat and mouse. *FASEB J* 37: e23299, 2023. doi:10.1096/fj.202301611R.
64. **Fuller KNZ**, **Thyfault JP.** Barriers in translating preclinical rodent exercise metabolism findings to human health. *J Appl Physiol (1985)* 130: 182–192, 2021. doi:10.1152/japplphysiol.00683.2020.
65. **Goutianos G**, **Tzioura A**, **Kyparos A**, **Paschalis V**, **Margaritelis NV**, **Veskoukis AS**, **Zafeiridis A**, **Dipla K**, **Nikolaidis MG**, **Vrabas IS.** The rat adequately reflects human responses to exercise in blood biochemical profile: a comparative study. *Physiol Rep* 3: e12293, 2015. doi:10.14814/phy2.12293. PMC]
66. **Gutierrez-Monreal MA**, **Harmsen JF**, **Schrauwen P**, **Esser KA.** Ticking for metabolic health: the skeletal-muscle clocks. *Obesity (Silver Spring)* 28, Suppl 1: S46–S54, 2020. doi:10.1002/oby.22826.
67. **Harfmann BD**, **Schroder EA**, **Kachman MT**, **Hodge BA**, **Zhang X**, **Esser KA.** Muscle-specific loss of Bmal1 leads to disrupted tissue glucose metabolism and systemic glucose homeostasis. *Skelet Muscle* 6: 12, 2016. doi:10.1186/s13395-016-0082-x.
68. **Sato S**, **Basse AL**, **Schönke M**, **Chen S**, **Samad M**, **Altıntaş A**, **Laker RC**, **Dalbram E**, **Barrès R**, **Baldi P**, **Treebak JT**, **Zierath JR**, **Sassone-Corsi P.** Time of exercise specifies the impact on muscle metabolic pathways and systemic energy homeostasis. *Cell Metab* 30: 92–110.e4, 2019. doi:10.1016/j.cmet.2019.03.013.
69. **Matsuzaki H**, **Daitoku H**, **Hatta M**, **Tanaka K**, **Fukamizu A.** Insulin-induced phosphorylation of FKHR (Foxo1) targets to proteasomal degradation. *Proc Natl Acad Sci USA* 100: 11285–11290, 2003. doi:10.1073/pnas.1934283100.
70. **Waddell DS**, **Baehr LM**, **van den Brandt J**, **Johnsen SA**, **Reichardt HM**, **Furlow JD**, **Bodine SC.** The glucocorticoid receptor and FOXO1 synergistically activate the skeletal muscle atrophy-associated MuRF1 gene. *Am J Physiol Endocrinol Physiol* 295: E785–E797, 2008. doi:10.1152/ajpendo.00646.2007.
71. **Breschi A**, **Gingeras TR**, **Guigó R.** Comparative transcriptomics in human and mouse. *Nat Rev Genet* 18: 425–440, 2017. doi:10.1038/nrg.2017.19.
72. **Jaura R**, **Yeh SY**, **Montanera KN**, **Ialongo A**, **Anwar Z**, **Lu Y**, **Puwakdandawa K**, **Rhee HS.** Extended intergenic DNA contributes to neuron-specific expression of neighboring genes in the mammalian nervous system. *Nat Commun* 13: 2733, 2022. doi:10.1038/s41467-022-30192-z.
73. **Yue F**, **Cheng Y**, **Breschi A**, **Vierstra J**, **Wu W**, **Ryba T**, **et al.** A comparative encyclopedia of DNA elements in the mouse genome. *Nature* 515: 355–364, 2014. doi:10.1038/nature13992.
74. **Graber TG**, **Borack MS**, **Reidy PT**, **Volpi E**, **Rasmussen BB.** Essential amino acid ingestion alters expression of genes associated with amino acid sensing, transport, and mTORC1 regulation in human skeletal muscle. *Nutr Metab (Lond)* 14: 35, 2017 [Erratum in *Nutr Metab (Lond)* 14: 39, 2017]. doi:10.1186/s12986-017-0187-1.

Downloaded from journals.physiology.org/journal/ajpregu (073.175.152.143) on October 8, 2026.